

Every rational cycle of the discrete-sine Hénon family

Abstract

We classify every rational periodic point of the conservative Hénon map $h_d(x, y) = (y, -x + s_d(y))$, where s_d is the integer-valued discrete-sine polynomial of any odd degree $d \geq 3$. The result gives the exact multiplicity of every ordinary primitive cycle, including both degree-dependent central phases and the unique growing boundary cycle when $d \equiv 1 \pmod{6}$. The total number of rational periodic points is $(3d^2 - 4d + 152)/3$, $d^2 + 6d + 24$, or $d^2 + 8d/3 + 14$, according as $d \equiv 5, 1, 3 \pmod{6}$. A nonarchimedean argument first proves integrality; a second-difference estimate then gives a uniform escape box. Inside its core, affine residue cells yield exact clipped multiplicities. Outside the core, a signed first-return graph with a complete endpoint complement provides the exhaustion. All formulas include the least admissible degrees. The resulting finite-cycle zeta is an immediate corollary of the ordinary cycle classification.

1 The family and its complete cycle census

An explicit Hénon map can have a large rational periodic set while leaving its exact cycle structure undetermined. For the discrete-sine family, the essential distinction is between the small-amplitude core and a narrow boundary on which new periods occur. We determine both pieces, with their actual point labels and ordinary iteration times.

For an odd integer $d = 2k + 1 \geq 3$, define

$$s_d(Y) = \sum_{j=0}^k (-1)^{k-j} \frac{Y \prod_{i=1}^j (Y^2 - i^2)}{(2j+1)!}, \quad h_d(x, y) = (y, -x + s_d(y)). \quad (1.1)$$

The empty product is 1. The degree of s_d is exactly d , and $h_d^{-1}(x, y) = (s_d(x) - y, x)$; thus h_d is a polynomial automorphism of $\mathbb{Q}^2$ with Jacobian determinant 1. Throughout, a primitive cycle means an orbit of least positive period, counted once under cyclic rotation. We do not identify a cycle with its negative or its reversal.

Put

$$R = \frac{d+1}{2} = 3q + s, \quad s \in \{0, 1, 2\}. \quad (1.2)$$

Here q is an integer quotient, not a finite-field cardinality. The allowed ranges are $q \geq 1$ when $s = 0, 1$, and $q \geq 0$ when $s = 2$. Let $C_n(d)$ count primitive rational cycles of length n .

Theorem 1.1 (Complete rational cycle classification). *For the map (1.1), every rational periodic point is integral and lies in $[-R - 2, R + 2]^2$. All primitive rational cycles are given by the following three additive tables. An omitted period has multiplicity zero; entries of the same period in different tables are added.*

Condition	Central cycles, written as period: multiplicity		
R odd	1 : 1,	5 : 2,	6 : 1
R even	1 : 1,	3 : 2,	10 : 1

Table 1: The central part, containing 17 points in either phase.

s	$Bulk\ C_4$	$Bulk\ C_{12}$	$Bulk\ C_{20}$
0	$q(q+1)$	$q(q-1)$	$q(q-1)$
1	$q(q+1)$	$q(q+1)$	$q(q-1)$
2	$q(q+1)$	$q(q+1)$	$q(q+1)$

Table 2: Generic core cycles, excluding the central part.

s	<i>Boundary cycles, written as period: multiplicity</i>		
0	$4 : 2q + 2,$	$14 : 2$	
1	$6 : 2,$	$16 : 2,$	$36 : q - 1, \quad (16q + 6) : 1$
2	$4 : 2,$	$6 : 1,$	$8 : 2q$

Table 3: All cycles meeting the boundary section.

The tables specify an explicit point classification, not only multiplicities. The positive-phase core consists exactly of the clipped residue rectangles in Table 8, including its central exceptions. The negative-phase point labels are obtained by the conjugacy of Proposition 5.1. The boundary cycles are the complete first-return families of Section 7; their intermediate points are recovered by iterating (1.1).

The word “bulk” below always refers to the generic core cycles in Table 2; it does not include the 17 central points. The word “boundary” refers to an orbit that meets the section, not to the assertion that all its points lie on the boundary of a square.

Corollary 1.2 (Point totals and ordinary return law). *The total number $P(d) = \# \text{Per}(h_d, \mathbb{Q})$ is*

$$P(d) = \begin{cases} 36q^2 - 20q + 53 = (3d^2 - 4d + 152)/3, & R = 3q, \quad d \equiv 5 \pmod{6}, \\ 36q^2 + 48q + 31 = d^2 + 6d + 24, & R = 3q + 1, \quad d \equiv 1 \pmod{6}, \\ 36q^2 + 52q + 31 = d^2 + 8d/3 + 14, & R = 3q + 2, \quad d \equiv 3 \pmod{6}. \end{cases} \quad (1.3)$$

For each $n \geq 1$, let $N_n(d) = \# \text{Fix}(h_d^n, \mathbb{Q})$. Then

$$N_n(d) = \sum_{\ell|n} \ell C_\ell(d), \quad Z_d(u) = \exp \left(\sum_{n \geq 1} \frac{N_n(d)}{n} u^n \right) = \prod_{\ell \geq 1} (1 - u^\ell)^{-C_\ell(d)}. \quad (1.4)$$

The product is finite, and the exponential identity is an identity of formal power series, as well as an analytic identity for $|u| < 1$.

The proof proceeds by integrality and escape, exact clipping of all affine core cells, and a signed first-return calculation on two boundary layers. Keeping the return signs is essential: a sign-negative return cycle of total time T produces ordinary period $2T$, not T .

2 Prior work and the scope of the increment

Kim, Krieger, Postolache and Szeto introduced the Hénon family used here, building on integer-valued polynomial constructions. Their Corollary 4.3 gives the rational/integer escape box, Theorem 5.1 gives the growing cycle, and Proposition 5.2 classifies the entire positive-phase lattice map, including all 17 central exceptional points [2]. These results, the polynomial family itself, and the existence

of many rational periodic points are prior work. We rederive the needed identities to make the argument complete.

Our increment is the exact all-radius clipping of that lattice classification together with the exhaustive signed boundary graph. This determines every multiplicity and excludes every additional rational cycle at every odd degree, including the first radius in each congruence class. The long-cycle argument below supplies its place and uniqueness in this graph, not a new claim of existence.

The comparison is specifically with arXiv:2412.01668v2. In that text, Theorem 4.4 gives at least 225 points for $d = 13$, while a subsequent exact-count remark gives 153. Formula (1.3) gives 271. We do not extend this observation to an unexamined final journal version or infer worldwide priority from a bounded source comparison. The arXiv version header is dated 8 July 2025; the served HTML separately displays an internal date of 24 August 2026. These are distinct metadata fields, not a reason to conflate versions [2].

Ingram's arithmetic study provides related nonarchimedean denominator mechanisms and rational-periodicity questions [1]. In particular, the computations in its Section 7 concern $(y, x + y^2 + b)$, whose sign differs from the conservative map in (1.1). We do not transfer a cycle classification across that sign change. The present theorem concerns only the exact factorial-product family (1.1), not shifted polynomials, all integer-valued polynomials, or arbitrary degree- d Hénon maps.

3 Integer values, rational integrality and escape

Define the six-periodic function $\sigma : \mathbb{Z} \rightarrow \{0, 1, -1\}$ by

$$(\sigma(0), \sigma(1), \dots, \sigma(5)) = (0, 1, 1, 0, -1, -1), \quad p = (-1)^{R-1}. \quad (3.1)$$

The polynomial s_d is odd. The following identities isolate all its values needed inside the eventual escape box.

Lemma 3.1 (Exact core and boundary values). *The polynomial s_d takes integer values on $\mathbb{Z}$. With $a \in \{0, 1, 2\}$ the residue $-R \bmod 3$ and $s = R \bmod 3$, one has*

$$s_d(n) = p\sigma(n) \quad (|n| \leq R), \quad s_d(R+1) = a, \quad s_d(R+2) = 2R + s. \quad (3.2)$$

Proof. Write

$$c_j(Y) = \frac{Y \prod_{i=1}^j (Y^2 - i^2)}{(2j+1)!}.$$

For an integer $n \geq 0$, $c_j(n) = \binom{n+j}{2j+1}$, with value zero when $j \geq n$. This proves integrality for $n \geq 0$; oddness proves it for $n < 0$. The following calculation takes place in formal power series, so each coefficient involves only finitely many summands:

$$\begin{aligned} \sum_{n \geq 0} \left(\sum_{j \geq 0} (-1)^j \binom{n+j}{2j+1} \right) z^n &= \sum_{j \geq 0} \frac{(-1)^j z^{j+1}}{(1-z)^{2j+2}} \\ &= \frac{z}{1-z+z^2}. \end{aligned}$$

The coefficient sequence on the right is $\sigma(n)$. Since $k = R - 1$, truncation gives $s_d(n) = p\sigma(n)$ for $0 \leq n \leq R$. At $n = R + 1$ the omitted tail is the single binomial coefficient 1; at $n = R + 2$ its two binomial coefficients are $2R + 2$ and 1. After their signs are included, the two truncated values are $1 + p\sigma(R + 1)$ and $2R + 1 + p\sigma(R + 2)$. Evaluating the six residues of R gives a and $2R + s$ respectively. Oddness supplies all negative values. $\square$

Lemma 3.2 (Uniform integer escape estimate). *For every odd $d \geq 3$,*

$$s_d(n) \geq 3n \quad (n \in \mathbb{Z}, n \geq R+3). \quad (3.3)$$

Consequently $|s_d(n)| \geq 3|n|$ for $|n| \geq R+3$.

Proof. Two applications of Pascal's identity give

$$c_j(n+1) - 2c_j(n) + c_j(n-1) = c_{j-1}(n) \quad (j \geq 1),$$

and $c_0(n) = n$ has zero second difference. Therefore

$$s_d(n+1) - 2s_d(n) + s_d(n-1) = s_{d-2}(n). \quad (3.4)$$

For $d = 3$ one has $R = 2$ and $s_3(n) = (n^3 - 7n)/6 \geq 3n$ for $n \geq 5$. Assume the assertion at degree $d-2$, whose core radius is $R-1$, and take $R \geq 3$. Put $D(n) = s_d(n) - s_d(n-1)$. By Lemma 3.1,

$$D(R+2) = 2R + s - a \geq 2R - 2 \geq 4.$$

For $n \geq R+2$, the induction hypothesis applies to $s_{d-2}(n)$; hence (3.4) gives $D(n+1) - D(n) = s_{d-2}(n) \geq 3n > 0$. Moreover,

$$\begin{aligned} s_d(R+3) &= 2s_d(R+2) - s_d(R+1) + s_{d-2}(R+2) \\ &\geq 4R - 2 + 3(R+2) = 7R + 4 \geq 3(R+3). \end{aligned}$$

All subsequent increments exceed 3, proving (3.3) by induction on n , and then on the odd degree. The absolute-value assertion follows by oddness. $\square$

Proposition 3.3 (Rational periodic points lie in the integer box). *Every rational periodic point of h_d belongs to*

$$B = [-R-2, R+2]^2 \cap \mathbb{Z}^2. \quad (3.5)$$

Proof. Write a periodic orbit as (y_{i-1}, y_i) , with indices taken cyclically. The coordinate recurrence is

$$y_{i-1} + y_{i+1} = s_d(y_i). \quad (3.6)$$

Fix a prime v and its usual nonarchimedean absolute value. If $|y|_v > 1$, then $|y^2 - i^2|_v = |y|_v^2$ for every integer i . The absolute values of the summands in $s_d(y)$ are therefore

$$|y|_v^{2j+1} |(2j+1)!|_v^{-1} \quad (0 \leq j \leq k).$$

They strictly increase with j : the quotient of consecutive values is $|y|_v^2 |(2j+2)(2j+3)|_v^{-1} > 1$. The ultrametric inequality with a unique largest summand yields

$$|s_d(y)|_v = |y|_v^d |d!|_v^{-1} > |y|_v.$$

If some orbit coordinate had norm larger than 1, choose one of maximal norm. Equation (3.6) would have right-hand norm larger than either term on the left, a contradiction. Every orbit coordinate is integral at every prime, hence lies in $\mathbb{Z}$.

Now let M be the maximal ordinary absolute value of the orbit coordinates. If $M \geq R+3$, Lemma 3.2 makes the right side of (3.6), at a maximizing coordinate, have absolute value at least $3M$; the left side has absolute value at most $2M$. Thus $M \leq R+2$. $\square$

In particular, a trajectory that leaves B cannot be periodic. This is the only meaning of “escape” required below: no assertion about two-sided divergence of every nonperiodic orbit is needed.

4 Clipped affine cells in the core

Let

$$B_0 = [-R, R]^2 \cap \mathbb{Z}^2, \quad g_p(x, y) = (y, -x + p\sigma(y)), \quad g = g_+.$$

By Lemma 3.1, h_d and g_p agree as long as their second coordinate remains in $[-R, R]$. We first study g on the whole lattice and then identify exactly which of its orbits stay in B_0 . This separates periodicity from the radius-dependent clipping.

Proposition 4.1 (Exact positive-phase cells). *Fix $(e, f) \in \{0, \dots, 5\}^2$. The corresponding row of Table 8 gives a generic period $\ell \in \{4, 12, 20\}$, four offsets, and all points of smaller period in that residue cell. The entire g -orbit of $(x, y) \equiv (e, f) \pmod{6}$ lies in B_0 if and only if*

$$-R + L_x \leq x \leq R + U_x, \quad -R + L_y \leq y \leq R + U_y. \quad (4.1)$$

Except at the indicated exceptional points, its least period is exactly ℓ . The central exceptions form the origin, two 5-cycles and one 6-cycle.

Proof. Within a fixed residue cell, write each coordinate as an affine expression in the free variables x, y . The residue of $ax + by + c$ is $ae + bf + c \pmod{6}$, so the exact affine iteration is

$$(X, Y) \mapsto (Y, -X + \sigma(Y \pmod{6})). \quad (4.2)$$

Table 8 gives all 36 resulting identities $g^\ell(x, y) = (x, y)$. Every intermediate coordinate is of the form $\pm x + c$ or $\pm y + c$. Intersecting the inequalities $-R \leq \pm x + c \leq R$ and their y analogues gives precisely (4.1). This is an intersection of affine inequalities, not a test at a chosen radius.

For clarity, the verification of any row consists of the following finite exact operations:

- (i) Start at the affine pair $(X_0, Y_0) = (x, y)$ and apply (4.2) through the listed value of ℓ .
- (ii) Check $(X_\ell, Y_\ell) = (x, y)$ as an affine identity and intersect all coordinate bounds through time $\ell - 1$.
- (iii) For each proper divisor t of ℓ , solve $X_t = x, Y_t = y$. Retain precisely the integral solutions with residues (e, f) .

Every consistent proper-divisor system in these rows has rank two; there is no consistent rank-one family to discard. The resulting solutions are exactly the points in the last column of the table. These operations, with (3.1), also give a direct reproducible verification of the table without external code. For example, in cell $(0, 0)$ the affine orbit is $(x, y), (y, -x), (-x, -y), (-y, x)$, giving period 4, zero clipping offsets, and only $(0, 0)$ as a proper-period point.

In actual point notation the exceptional cycles are the fixed point $(0, 0)$, the cycle

$$(0, 2) \rightarrow (2, 1) \rightarrow (1, -1) \rightarrow (-1, -2) \rightarrow (-2, 0) \rightarrow (0, 2) \quad (4.3)$$

and its negative, and

$$(0, 1) \rightarrow (1, 1) \rightarrow (1, 0) \rightarrow (0, -1) \rightarrow (-1, -1) \rightarrow (-1, 0) \rightarrow (0, 1). \quad (4.4)$$

All 17 points lie in $[-2, 2]^2$, and thus occur for every $R \geq 2$. The proper-divisor equations exclude every other exceptional point. $\square$

We next count the clipped rectangles exactly. Put $R = 6m + r$ with $0 \leq r \leq 5$. For a single coordinate of residue e and offsets (L, U) , write the coordinate as $e + 6j$. The number of allowed

integers is

$$\begin{aligned} A_{r,e}^{L,U}(m) &= \left\lfloor \frac{R+U-e}{6} \right\rfloor - \left\lfloor \frac{-R+L-e}{6} \right\rfloor + 1 \\ &= 2m + \left\lfloor \frac{r+U-e}{6} \right\rfloor - \left\lfloor \frac{-r+L-e}{6} \right\rfloor + 1. \end{aligned} \quad (4.5)$$

For the rows in Table 8, these values are nonnegative already at the least permitted m : it is 1 for $r = 0, 1$ and 0 for the other residues. No positive-part correction is omitted.

Let $\eta_{e,f}$ be 1 if the last column of that row contains a point, and 0 otherwise. All such points satisfy the clipping inequalities at every $R \geq 2$. Thus the number of generic cycles of a given period ℓ is exactly

$$\frac{1}{\ell} \sum_{\substack{(e,f) \\ \text{row period } \ell}} \left(A_{r,e}^{L_x,U_x}(m) A_{r,f}^{L_y,U_y}(m) - \eta_{e,f} \right). \quad (4.6)$$

Each generic orbit is counted at each of its ℓ points, possibly in different residue cells; division by ℓ is therefore necessary. Expanding the products gives Table 4.

r	Bulk C_4	Bulk C_{12}	Bulk C_{20}
0	$4m^2 + 2m$	$4m^2 - 2m$	$4m^2 - 2m$
1	$4m^2 + 2m$	$4m^2 + 2m$	$4m^2 - 2m$
2	$4m^2 + 2m$	$4m^2 + 2m$	$4m^2 + 2m$
3	$4m^2 + 6m + 2$	$4m^2 + 2m$	$4m^2 + 2m$
4	$4m^2 + 6m + 2$	$4m^2 + 6m + 2$	$4m^2 + 2m$
5	$4m^2 + 6m + 2$	$4m^2 + 6m + 2$	$4m^2 + 6m + 2$

Table 4: Exact rectangle counts in the six radius classes.

Since $q = 2m + \lfloor r/3 \rfloor$, substitution gives exactly Table 2. This derivation uses the affine cells and arithmetic progressions for arbitrary m , not polynomial fitting to finite degree samples.

5 The negative phase and actual point labels

The sign p in (3.1) changes with the parity of R . It changes the central cycle structure, but not the generic bulk multiplicities. To keep the ordinary clock and the point coordinates correct, introduce

$$S(x, y) = (-x, -y), \quad C(x, y) = (-x, y).$$

Proposition 5.1 (Phase transfer). *The identities*

$$Sg = gS, \quad g_- = C(Sg)C \quad (5.1)$$

hold on $\mathbb{Z}^2$. Let $\mathcal{K}_+$ be the set described by the clipped cells of Proposition 4.1. The set of points whose entire g_- orbit lies in B_0 is $C\mathcal{K}_+$. For $z \in \mathcal{K}_+$, its actual negative-phase orbit is

$$g_-^n(Cz) = CS^n g^n z. \quad (5.2)$$

All generic periods 4, 12, 20 and their multiplicities are unchanged. The positive central cycles $1 : 1, 5 : 2, 6 : 1$ become $1 : 1, 3 : 2, 10 : 1$ in the negative phase.

Proof. Oddness of σ proves $Sg = gS$; direct substitution proves the second identity. Both S and C preserve B_0 . Since $(Sg)^n = S^n g^n$, a point has its entire Sg orbit in B_0 if and only if it has its entire g orbit there. Conjugating by C proves the set statement and (5.2).

Suppose that z has g -period $\ell \in \{4, 12, 20\}$. Because ℓ is even, $(Sg)^\ell z = z$. If $(Sg)^n z = z$ with n even, then $g^n z = z$, so $\ell \mid n$. If n is odd, then $g^n z = -z$, which implies $g^{2n} z = z$ and hence $\ell \mid 2n$. This is impossible because $4 \mid \ell$ but n is odd. Thus its least Sg period is exactly ℓ . The number of points of each generic period, and consequently the cycle multiplicity, does not change.

Negation exchanges the two cycles in (4.3), so Sg joins their ten points into one 10-cycle. On the cycle (4.4), negation equals g^3 ; therefore $Sg = g^4$ there, producing two 3-cycles. The origin is still fixed. Applying C to these orbit labels yields the claimed negative-phase cycles for g_- . $\square$

For example, a negative-phase central 10-cycle is obtained by applying CS^n to the n th point of (4.3), with the latter point read modulo 5, for $0 \leq n < 10$. This is an explicit reconstruction of ten actual points. Directly reusing the positive-phase labels without C would not be the stated conjugacy. Because $p = +1$ precisely when R is odd, Proposition 5.1 proves Table 1 and the bulk part of Theorem 1.1 for the actual map h_d .

6 A signed first-return section with all endpoints

Every periodic orbit not wholly in B_0 has a coordinate of absolute value $R + 1$ or $R + 2$, by Proposition 3.3. Hence it meets the section

$$A = \{(x, y) \in B : |y| \in \{R + 1, R + 2\}\}. \quad (6.1)$$

The map h_d commutes with S . For a pair of opposite section points, use the unique representative with positive second coordinate:

$$[x, \lambda] = (x, R + \lambda), \quad \lambda \in \{1, 2\}.$$

This notation is only a proof device. Signs on the return edges retain the ordinary, nonquotiented dynamics.

A return edge

$$[x, \lambda] \xrightarrow{\tau, \varepsilon} [x', \lambda']$$

means that τ is the first positive return time to A and $h_d^\tau(x, R + \lambda) = \varepsilon(x', R + \lambda')$, where $\varepsilon \in \{+1, -1\}$. If a trajectory leaves B instead, it is discarded from periodic classification.

Lemma 6.1 (Ordinary lift of a signed return cycle). *Let a primitive normalized section cycle have total time T and edge-sign product ε . If $\varepsilon = +1$, it lifts to two distinct primitive h_d -cycles of length T . If $\varepsilon = -1$, it lifts to one primitive h_d -cycle of length $2T$.*

Proof. Following the return edges once sends a representative z to εz . Since A has no point fixed by S , a negative sign requires two traversals to return to z , whereas a positive sign requires one. An earlier ordinary return would imply an earlier repeat of the normalized section point, contradicting its primitive return cycle and the first-return property. In the positive case the lifts through z and $-z$ cannot coincide: such a coincidence would give a negative return to the same normalized starting point before the positive traversal closed. Thus the stated periods and cycle counts are ordinary ones. $\square$

We use the endpoint labels

$$L_c = [-R + c, 1], \quad U_c = [R + c, 1], \quad V_c = [R + c, 2]. \quad (6.2)$$

Let $r = R \bmod 6$, $\kappa = 3\lfloor r/3 \rfloor$, and $e = (x - \kappa) \bmod 6$. The phase adjustment in e makes it possible to combine pairs of radius classes in the next table.

Lemma 6.2 (All generic inner returns). *For $[x, 1]$ with $-R + L \leq x \leq R + U$ and residue e as above, Table 5 gives its first return. Every edge has sign -1 and normalized target $[x + \delta, \lambda']$.*

s	e	L	U	τ	δ	λ'
0	0, 3	0	0	2	0	1
0	1	2	0	10	-2	1
0	2	4	0	14	-4	1
0	4, 5	0	0	2	0	2
1	0	4	2	6	-4	1
1	1	3	-1	18	0	1
1	2, 5	2	2	2	-2	1
1	3, 4	2	2	2	-2	2
2	0, 5	1	1	2	-1	2
2	1, 4	1	1	2	-1	1
2	2	1	-3	10	3	1
2	3	1	-1	6	1	1

Table 5: Generic inner first returns; all signs are negative.

Proof. Keep R and x as free affine variables, with their residues fixed. Starting from $(x, R + 1)$, the first step is $(R + 1, a - x)$. Use (3.2) at the initial boundary coordinate and $p\sigma(y)$ at each subsequent core coordinate. Record the inequality $-R \leq y \leq R$ at every intermediate second coordinate. The exact intersection of its x inequalities is the displayed interval. All other coordinates encountered before the return are of the form $\pm R + c$ or constants; their core inequalities hold on the relevant radius progression already at its least radius.

More explicitly, for each of the six choices of r and each $e = 0, \dots, 5$, substitute $x \equiv e + \kappa \pmod{6}$ into the affine recurrence. Iterate until the second coordinate is identically $\pm(R + \lambda')$. In all 36 cases this occurs at the listed τ , the last pair is $-(x + \delta, R + \lambda')$, and the intersected bounds have the stated offsets. At each earlier time the recorded second coordinate is in $[-R, R]$, so this is a first return. These steps are a finite identity check in two affine variables; they verify the table for every R in the progression simultaneously. The least radii for $r = 0, 1, \dots, 5$ are respectively 6, 7, 2, 3, 4, 5, which also cover the smallest allowed degrees. $\square$

For an outer state $[x, 2]$, the next second coordinate is $2R + s - x$. Thus it leaves B immediately if

$$x < R + s - 2. \quad (6.3)$$

Every other outer representative is one of the finitely many V_c with $s - 2 \leq c \leq 2$. There is no unexamined long outer strip.

The inner endpoint list is likewise an exact complement, not a bounded search near corners. For any row of Table 5, the complement of its generic interval in $[-R - 2, R + 2]$ is

$$x = -R + c, \quad -2 \leq c \leq L - 1, \quad \text{or} \quad x = R + c, \quad U + 1 \leq c \leq 2, \quad (6.4)$$

filtered by that row's residue e . The generic real interval is nonempty at the least allowed radius, since $2R + U - L \geq 0$ there. The two complementary intervals are consequently disjoint. Applying (6.4) gives exactly six exceptional inner states in each radius class. Together with the surviving outer states they give the complete endpoint table below.

Lemma 6.3 (Complete endpoint returns and escapes). *All section representatives outside the generic inner strips either escape immediately by (6.3) or occur as a start in Table 6. The table gives all nonescaping first returns. The symbol E denotes an escape path, whose precise short transient is specified after the table.*

Table 6: Complete exceptional inner and surviving outer rules. The paired L row for $s = 1$ represents two states.

s	Start	First normalized target	Time	Sign
0	L_{-2}	V_1	1	+
0	L_{-1}	U_1	1	+
0	L_1	U_{-1}	7	−
0	L_2	U_{-2}	9	+
0	V_{-2}	V_2	1	+
0	V_{-1}	U_2	1	+
0	V_0	E	—	—
0	U_1	L_{-1}	1	−
0	V_1	L_1	2	−
0	U_2	E	—	—
0	V_2	L_2	2	−
1	L_{-2}, L_{-1}	E	—	—
1	L_0	V_1	1	+
1	L_1	U_1	1	+
1	L_2	U_0	3	−
1	V_{-1}	V_2	1	+
1	U_0	L_2	13	−
1	V_0	U_2	1	+
1	V_1	L_0	2	−
1	V_2	E	—	—
2	L_{-2}	E	—	—
2	L_{-1}	V_1	1	+
2	L_0	U_1	1	+
2	U_0	L_1	5	−
2	V_0	V_2	1	+
2	U_1	L_0	3	+
2	V_1	U_2	1	+
2	U_2	L_{-1}	1	−
2	V_2	E	—	—

Proof. The complement calculation (6.4) and the outer threshold prove that the start list is exhaustive. Each row then follows from the recurrence (3.6), using (3.2) and oddness at a boundary coordinate. Here R alone is free. Reduce each affine coordinate modulo 6 on its fixed radius progression and continue until the first return or escape. The intervening core inequalities are linear in R ; checking their direction and their value at the least allowed radius proves them throughout the progression.

For the longest endpoint return, namely $s = 1$ and $U_0 = (R, R + 1)$, the second-coordinate list from time 0 through time 13 is

$$\begin{aligned} & R + 1, -R + 2, -R, R - 3, R - 1, -R + 3, -R + 2, \\ & R - 2, R - 3, -R + 1, -R + 3, R, R - 2, -R - 1. \end{aligned}$$

The final pair is $-L_2$, and all intermediate second coordinates lie in $[-R, R]$ already at $R = 4$. This illustrates both the negative sign and the first-return check.

For precision about the E entries, when $s = 0$ the paths are

$$V_0 \xrightarrow{2,-} [-R, 2] \longrightarrow E, \quad U_2 \xrightarrow{1,-} [-R - 1, 2] \longrightarrow E.$$

Their displayed outer targets satisfy (6.3). When $s = 1$, the path is $V_2 \xrightarrow{2,-} [-R + 1, 2] \longrightarrow E$. When $s = 2$, V_2 leaves B in time 2. Every L state marked E leaves B in time 1. Thus the compression of these transient paths to E does not discard any candidate cycle. $\square$

Lemma 6.4 (No endpoint aliases at small radii). *For every allowed R , the exceptional inner labels in Table 6 are pairwise distinct and lie outside all generic strips. The outer labels are distinct from the inner labels.*

Proof. For $s = 0$, the largest left inner coordinate is $-R + 2$ and the smallest right one is $R + 1$, with $R \geq 3$. For $s = 1$, they are $-R + 2$ and R , with $R \geq 4$. For $s = 2$, they are $-R$ and R , with $R \geq 2$. Each gap is strictly positive already at the least radius and increases with R . Different offsets on the same side differ by a nonzero constant. Outer labels have second coordinate $R + 2$ rather than $R + 1$. Finally, (6.4) separates every inner endpoint from every generic strip of its residue, and different residues cannot coincide. $\square$

7 Closing and exhausting the boundary graph

We now use the return rules to assign every normalized section state to a cycle or an escape path. In the strings below, $[x, 1]$ is occasionally abbreviated by its coordinate x when the layer is unambiguous. Ordinary cycle multiplicities always come from Lemma 6.1, not from counting normalized cycles.

7.1 Radius $R = 3q$

The $e = 0, 3$ strip consists precisely of the multiples of 3 in $[-R, R]$. There are $2q + 1$ such coordinates. Each is a normalized fixed point of time 2 and negative sign, hence gives one ordinary 4-cycle.

The endpoint cycle

$$L_{-1} \xrightarrow{1,+} U_1 \xrightarrow{1,-} L_{-1}$$

has total time 2 and negative sign, giving one more 4-cycle. The other periodic endpoint string is

$$L_2 \xrightarrow{9,+} U_{-2} \xrightarrow{2,-} V_{-2} \xrightarrow{1,+} V_2 \xrightarrow{2,-} L_2. \quad (7.1)$$

The U_{-2} edge is the $e = 4$ strip rule. Its total time is 14 and its sign is positive, giving two 14-cycles. The normalized nodes in these cycles are distinct: the inner endpoints are separated by Lemma 6.4, the outer layer is different, and its two offsets differ by 4.

To prove exhaustion, consider each other strip residue. A generic $e = 1$ state first goes to $e = 5$ by $x \mapsto x - 2$, then to the outer layer. Its outer coordinate is at most $R - 7 < R - 2$. A generic

$e = 2$ state goes to $e = 4$ by $x \mapsto x - 4$, then to an outer coordinate at most $R - 8 < R - 2$. Both paths escape by (6.3). Of the generic $e = 4, 5$ states that go directly to the outer layer, only U_{-2} and U_{-1} reach an outer coordinate at least $R - 2$. The first is in (7.1), while the second follows

$$U_{-1} \longrightarrow V_{-1} \longrightarrow U_2 \longrightarrow E.$$

The unused left endpoint path is $L_{-2} \rightarrow V_1 \rightarrow L_1 \rightarrow U_{-1}$, and V_0 escapes. This accounts for every start in Tables 5 and 6. Thus the boundary multiplicities are

$$4 : 2q + 2, \quad 14 : 2.$$

7.2 Radius $R = 3q + 1$

The stationary $e = 1$ strip contains exactly $q - 1$ coordinates. They are congruent to R modulo 6 and run from $-R + 8$ to $R - 6$ in increments of 6. At $q = 1$ there is no such coordinate. Each stationary state has return time 18 and negative sign, giving one ordinary 36-cycle.

The two normalized cycles

$$L_0 \xrightarrow{1,+} V_1 \xrightarrow{2,-} L_0, \quad V_0 \xrightarrow{1,+} U_2 \xrightarrow{2,-} V_0$$

each have total time 3 and negative sign, and give two ordinary 6-cycles in total. The U_2 edge here is a generic strip edge, not an omitted exceptional start. The endpoint cycle

$$L_2 \xrightarrow{3,-} U_0 \xrightarrow{13,-} L_2$$

has time 16 and positive sign, hence gives two 16-cycles. All these nodes are separated by their layer, offsets and Lemma 6.4.

Proposition 7.1 (The full growing-cycle string). *For $R = 3q + 1$, there is exactly one ordinary boundary cycle of length $16q + 6$. Its normalized section cycle consists of the initial edge $L_1 \xrightarrow{1,+} U_1$ and the interleaved strings defined below. No generic $e = 0$ or $e = 2$ state lies outside this cycle.*

Proof. For $0 \leq j \leq q$, put

$$a_j = R + 1 - 6j.$$

These are exactly the generic $e = 2$ coordinates, with $a_0 = R + 1$ and $a_q = -R + 3$. Their time-2 negative edges go to $a_j - 2$. For $j < q$, this is a generic $e = 0$ coordinate; its time-6 negative edge goes to a_{j+1} . For $j = q$, the target is $a_q - 2 = -R + 1$, namely L_1 . Thus the normalized cycle is

$$\begin{aligned} L_1 \xrightarrow{1,+} [a_0, 1] \xrightarrow{2,-} [a_0 - 2, 1] \xrightarrow{6,-} [a_1, 1] \xrightarrow{2,-} \dots \\ \dots \xrightarrow{6,-} [a_q, 1] \xrightarrow{2,-} L_1. \end{aligned}$$

It has $q + 1$ time-2 edges, q time-6 edges, and one time-1 edge. Its total time and sign are

$$T = 1 + 2(q + 1) + 6q = 8q + 3, \quad \varepsilon = (-1)^{2q+1} = -1.$$

Lemma 6.1 gives one ordinary cycle of length $2T = 16q + 6$. The strict decrease by 6 prevents repetition within either string; their different residues separate the two strings. The endpoint L_1 is outside the generic strip by (6.4). This proves primitivity.

Finally, the entire generic $e = 0$ progression runs from $-R + 7$ to $R - 1$, and the entire generic $e = 2$ progression runs from $-R + 3$ to $R + 1$, each in steps of 6. They are precisely the nodes just listed. This proves the asserted coverage; exhaustion of the other residues below then proves uniqueness in the full graph. $\square$

Every generic $e = 5$ state goes to $e = 3$, then to an outer coordinate at most $R - 6 < R - 1$, and escapes. Among generic $e = 3, 4$ states that go directly to the outer layer, only U_2 reaches an outer coordinate at least $R - 1$; it was already assigned to the second 6-cycle. The remaining endpoint path is $V_{-1} \rightarrow V_2 \rightarrow E$, and L_{-2}, L_{-1} escape immediately. The stationary strip, the growing string, these escape paths, and the displayed endpoint cycles exhaust every section state. The boundary multiplicities are therefore

$$6 : 2, \quad 16 : 2, \quad 36 : q - 1, \quad (16q + 6) : 1.$$

7.3 Radius $R = 3q + 2$

For a generic $e = 3$ coordinate x , the target $x + 1$ is generic of residue $e = 4$ and returns to x . The two edges have times 6, 2 and signs $-, -$. Hence each pair gives two ordinary 8-cycles. The allowed $e = 3$ coordinates are

$$-R + 5, -R + 11, \dots, R - 5,$$

giving q pairs; at $q = 0$ the list is empty. Their partners account for all generic $e = 4$ coordinates. The residues are distinct, so these normalized cycles are primitive.

The endpoint cycles are

$$L_{-1} \xrightarrow{1,+} V_1 \xrightarrow{1,+} U_2 \xrightarrow{1,-} L_{-1}, \quad L_0 \xrightarrow{1,+} U_1 \xrightarrow{3,+} L_0.$$

Their total times and signs are $(3, -)$ and $(4, +)$, giving one ordinary 6-cycle and two ordinary 4-cycles. Their nodes are distinct even at $R = 2$, by Lemma 6.4 and their different layers.

For exhaustion, a generic $e = 1$ state goes to $e = 0$ and then to an outer coordinate at most $R - 3 < R$. A generic $e = 2$ state goes to $e = 5$ and then to an outer coordinate at most $R - 4 < R$. Direct outer returns from generic $e = 0, 5$ also have outer coordinate below R . All these paths escape by (6.3). The unused endpoint paths are

$$U_0 \longrightarrow L_1 \longrightarrow E, \quad V_0 \longrightarrow V_2 \longrightarrow E, \quad L_{-2} \longrightarrow E.$$

Here L_1 is a generic $e = 5$ state; the arrow from it to E allows its transient outer return before escape. Every section state is assigned, yielding precisely

$$4 : 2, \quad 6 : 1, \quad 8 : 2q.$$

8 Exhaustiveness, least radii and the return law

Completion of the proof of Theorem 1.1. Proposition 3.3 places every rational periodic orbit in B . If it avoids A , it lies wholly in B_0 . Indeed, a point outside B_0 has a coordinate of magnitude $R + 1$ or $R + 2$, and that coordinate becomes a second coordinate on the same periodic orbit. The core orbit is therefore exactly one of those in Propositions 4.1 and 5.1. Conversely, every clipped core orbit agrees with h_d at every step and is an actual periodic orbit of that map.

If a periodic orbit meets A , its normalized section points follow the first-return rules of Section 6. The inner section is the disjoint union of the generic residue strips and the endpoint complement (6.4). Every other outer point escapes in one step. The three cases in Section 7 assign every possible start to a primitive normalized cycle or an escape path. Lifting by Lemma 6.1 therefore gives every remaining ordinary cycle exactly once. The explicit recurrence reconstructs all points between its section nodes. This proves the classification and all three additive tables. $\square$

All multiplicities in the tables are nonnegative on the allowed ranges. The growing period has $q \geq 1$, hence is at least 22; it never coincides with 36 because $16q + 6 = 36$ has no integer solution. It also differs from every other listed period. The repeated appearances of periods 4 and 6 in distinct tables represent disjoint core and boundary cycles and must be added. In particular, the three least radii give the following direct substitutions into the uniform formulas.

R	Central	Generic bulk	Boundary	$P(d)$
2	1 : 1, 3 : 2, 10 : 1	none	4 : 2, 6 : 1	31
3	1 : 1, 5 : 2, 6 : 1	4 : 2	4 : 4, 14 : 2	69
4	1 : 1, 3 : 2, 10 : 1	4 : 2, 12 : 2	6 : 2, 16 : 2, 22 : 1	115

Table 7: The least radii. Entries have the same period: multiplicity convention as the main theorem.

These are not separate numerical degree-graph computations. The empty 8-cycle family at $R = 2$ and empty 36-cycle family at $R = 4$ were already treated as empty arithmetic progressions. All affine inequalities and endpoint noncoincidences were proved at the least admissible radius and throughout its progression.

Proof of Corollary 1.2. There are always 17 central points. Multiplying the generic cycle counts by their lengths gives the respective bulk totals

$$36q^2 - 28q, \quad 36q^2 - 4q, \quad 36q^2 + 36q.$$

Similarly, the boundary totals for $s = 0, 1, 2$ are

$$8q + 36, \quad 52q + 14, \quad 16q + 14.$$

For example, in the middle case the second total is $2 \cdot 6 + 2 \cdot 16 + 36(q - 1) + (16q + 6) = 52q + 14$. Adding the central 17 gives (1.3); substituting $d = 2R - 1$ gives its degree forms.

An ordinary primitive ℓ -cycle contributes all its ℓ points to $\text{Fix}(h_d^n, \mathbb{Q})$ exactly when $\ell \mid n$. This proves the formula for $N_n(d)$. Consequently

$$\begin{aligned} \sum_{n \geq 1} \frac{N_n(d)}{n} u^n &= \sum_{\ell \geq 1} C_\ell(d) \sum_{j \geq 1} \frac{u^{\ell j}}{j} \\ &= - \sum_{\ell \geq 1} C_\ell(d) \log(1 - u^\ell). \end{aligned}$$

There are only finitely many nonzero $C_\ell(d)$, so this is valid formally and for $|u| < 1$. Exponentiation proves (1.4). $\square$

Boundaries of the result. The proof classifies all of $\mathbb{Q}^2$ for the exact family (1.1) and its ordinary clock. It does not classify nonperiodic orbits, complex periodic points, shifted families or arbitrary rational Hénon maps. The rational function Z_d records a finite source cycle permutation. It does not establish target Euler factors, a root number, automorphy, a target zero-divisor correspondence or a Hilbert–Pólya realization. The three radius classes are cases of one family theorem, and the zeta identity is its corollary, not a separate classification.

A The complete positive-phase affine table

The table uses the rectangle convention (4.1). Each row asserts an affine identity $g^\ell(x, y) = (x, y)$ on its entire residue cell; its last column lists every point in that cell having proper-divisor period. A dash means there is no such point. The verification procedure in Proposition 4.1 requires only the six values in (3.1), affine substitution through at most 20 steps, and two-variable linear equations. Thus the table is a finite proof table included in the article, not a reference to an unprovided degree census.

Table 8: All 36 positive-phase core cells and their central exceptions.

(e, f)	ℓ	(L_x, U_x)	(L_y, U_y)	Exceptional point: period
(0, 0)	4	(0, 0)	(0, 0)	(0, 0) : 1
(0, 1)	12	(1, -1)	(2, 0)	(0, 1) : 6
(0, 2)	20	(2, -2)	(4, 0)	(0, 2) : 5
(0, 3)	4	(0, 0)	(0, 0)	—
(0, 4)	20	(2, -2)	(0, -4)	(0, -2) : 5
(0, 5)	12	(1, -1)	(0, -2)	(0, -1) : 6
(1, 0)	12	(2, 0)	(1, -1)	(1, 0) : 6
(1, 1)	12	(2, 0)	(2, 0)	(1, 1) : 6
(1, 2)	20	(3, -1)	(4, 0)	(1, 2) : 5
(1, 3)	20	(0, -4)	(2, -2)	—
(1, 4)	20	(0, -4)	(3, -1)	—
(1, 5)	20	(3, -1)	(1, -3)	(1, -1) : 5
(2, 0)	20	(4, 0)	(2, -2)	(2, 0) : 5
(2, 1)	20	(4, 0)	(3, -1)	(2, 1) : 5
(2, 2)	20	(1, -3)	(1, -3)	—
(2, 3)	12	(0, -2)	(1, -1)	—
(2, 4)	12	(0, -2)	(2, 0)	—
(2, 5)	20	(1, -3)	(4, 0)	—
(3, 0)	4	(0, 0)	(0, 0)	—
(3, 1)	20	(2, -2)	(0, -4)	—
(3, 2)	12	(1, -1)	(0, -2)	—
(3, 3)	4	(0, 0)	(0, 0)	—
(3, 4)	12	(1, -1)	(2, 0)	—
(3, 5)	20	(2, -2)	(4, 0)	—
(4, 0)	20	(0, -4)	(2, -2)	(-2, 0) : 5
(4, 1)	20	(3, -1)	(0, -4)	—
(4, 2)	12	(2, 0)	(0, -2)	—
(4, 3)	12	(2, 0)	(1, -1)	—
(4, 4)	20	(3, -1)	(3, -1)	—
(4, 5)	20	(0, -4)	(1, -3)	(-2, -1) : 5
(5, 0)	12	(0, -2)	(1, -1)	(-1, 0) : 6
(5, 1)	20	(1, -3)	(3, -1)	(-1, 1) : 5
(5, 2)	20	(4, 0)	(1, -3)	—
(5, 3)	20	(4, 0)	(2, -2)	—
(5, 4)	20	(1, -3)	(0, -4)	(-1, -2) : 5
(5, 5)	12	(0, -2)	(0, -2)	(-1, -1) : 6

References

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